

David Harris

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Skills

Hardware: Verilog, PCB Design, Digital & Analog Circuits, SolidWorks

Programming: C, Python, Java, C++, Javascript, Assembly, React, Docker

Tools: Quartus, Git, VS Code, MATLAB, SolidWorks

Experience & Leadership

Software Developer / Contributor, Menace AIO (Startup) – Arlington, VA June 2022 – Present

- Debugged and improved a desktop automation application, identifying front-end software issues and implementing fixes to improve reliability and functionality for 200+ users.
- Developed and refined GUI components, improving application usability and user interaction.

Preschool Assistant/Teacher, St. Kathrine Greek Orthodox Church – Falls Church, VA June 2022 – Present

- Entrusted to independently teach and supervise groups of ~10 preschool students, leading English instruction and daily activities while maintaining a safe structured classroom environment and working with other teachers.

Event Manager + Interim President, Delta Kappa Epsilon Jan 2021 – May 2022

- **Event Manager:** Executed 25+ organization wide events for 50+ members by coordinating logistics, planning timelines, managing a five figure budget, and leading event operations.
- **Interim President:** Assumed presidential leadership on short notice, overseeing operations for 50+ members while coordinating with and traveling to meet with international leadership while also leading day-to-day operations.

Founder, Sneaker Resale Business June 2014 – July 2014

- Generated \$15k+ in profit through sourcing, pricing, inventory management and online sales.
- Managed AWS EC2 infrastructure for automated purchasing systems, configuring linux servers, proxies and concurrent processes.

Projects

Interactive Circuit Simulation & Algorithm Visualizer name.com/projectdemo

- Developed a React/TypeScript web app for building and simulation RC/RLC circuits, using graph based data structures to represent circuit topology and visualize voltage/current waveforms.
- Implemented graph traversal algorithms and numerical circuit analysis, with an interactive interface for modifying components and observing real time simulation results; containerized with Docker.

Custom 16 bit processor & transaction processing system | Verilog, Assembly, FPGA Design

- Designed and implemented a custom 16 bit processor in Verilog, including a function unit, instruction decoder, opcode decoding, and control signal generation for arithmetic, logic, shift and immediate operations.
- Developed an assembly based transaction processor utilizing memory mapped I/O, branching, bitwise decryption and data validation.
- Tested and Verified processor and software functionality through simulation, debugging, timing analysis and waveform verifications.

Arduino Based MIDI & Drum Pad Controller for Ableton Live

- Designed and programmed a custom MIDI controller using an arduino Mega with analog potentiometers, push button drum pads and serial communication to provide real time control of abletons live effects and midi triggers.
- Programmed embedded C/C++ firmware utilizing ADC, serial communication, and MIDI mapping to convert physical inputs into low latency digital audio controls.

Education

School – Pursuing BS in Electrical Engineering with a minor in computer science - Present

Relevant Courses: Embedded Systems, Quantum Engineering, Machine Learning, Circuits and Devices, Digital Systems, Data Structures & Algorithms, Computational Engineering.